

TAPACHULA, Mexico

WMO: 769030

Lat: 14.8872N

Lon: 92.2964W

Elev: 118

StdP: 99.92

Time Zone: -6.00 (MEX)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	20.1	20.9	16.0	11.5	24.8	17.1	12.4	24.9	12.9	32.0	9.4	26.2	1.5	40	0.305

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.1	35.6	26.1	34.9	26.1	34.3	26.0	28.4	33.1	27.9	32.8	27.5	32.4	3.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.1	23.2	31.9	26.5	22.4	31.4	26.1	21.8	30.9	92.3	33.1	89.8	32.9	87.8	32.5	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.4	6.4	4.8	13.9	37.3	3.0	1.1	11.8	38.1	10.0	38.7	8.4	39.3	6.2	40.0
			14.0	29.4	1.5	1.1	12.9	30.2	12.1	30.8	11.2	31.4	10.2	32.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	27.1	28.0	28.8	29.4	28.9	27.9	28.0	28.0	27.5	27.5	27.6	27.1
	DBStd	1.31	1.08	1.08	1.06	1.21	1.13	1.29	1.13	1.10	1.17	1.10	0.98	1.00
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6561	530	504	583	581	585	537	557	557	525	544	529	528
	CDD18.3	3519	272	271	325	331	327	287	299	299	275	286	279	270
	CDH23.3	35168	2523	2817	3655	3870	3583	2771	2937	2852	2377	2603	2724	2456
	CDH26.7	15149	1131	1328	1751	1857	1578	1102	1236	1149	859	993	1111	1054
(14) Wind	WSAvg	1.8	1.7	2.0	2.2	2.0	1.9	1.8	1.7	1.7	1.6	1.6	1.6	1.4
(15) Precipitation	PrecAvg	2356	9	3	24	89	260	368	309	344	422	318	96	8
	PrecMax	3089	70	33	124	334	585	808	730	596	791	695	371	47
	PrecMin	1645	0	0	0	0	30	101	73	172	232	20	0	0
	PrecStd	365	17	6	35	65	116	145	128	102	147	145	100	11
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.6	35.4	36.1	36.7	35.9	34.8	34.9	34.7	34.1	34.3	34.2	34.4
		MCWB	24.9	25.1	25.8	26.4	26.3	27.6	27.4	27.5	27.2	27.0	26.7	25.3
	2%	DB	33.6	34.5	35.2	35.6	34.7	33.6	34.0	33.8	33.1	33.2	33.3	33.3
		MCWB	24.8	24.8	25.5	26.2	26.7	27.2	27.2	27.3	26.9	26.7	26.2	25.3
	5%	DB	32.9	33.8	34.5	34.8	33.8	32.8	33.1	32.9	32.1	32.4	32.6	32.4
		MCWB	24.6	24.7	25.3	26.1	26.7	26.9	26.9	26.9	26.5	26.6	26.0	25.3
	10%	DB	32.0	33.0	33.6	33.9	32.8	31.9	32.2	32.0	31.2	31.5	31.8	31.6
		MCWB	24.3	24.4	25.2	25.9	26.7	26.5	26.6	26.5	26.2	26.3	25.8	25.2
	0.4%	WB	27.1	27.3	28.1	29.2	28.8	28.7	28.8	28.6	28.1	28.3	27.9	27.2
		MCDB	32.5	33.1	33.8	33.6	33.1	33.3	33.2	33.3	32.8	32.5	32.7	31.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.1	26.3	26.9	27.7	27.9	27.7	27.8	27.8	27.3	27.6	27.2	26.6
		MCDB	31.9	32.5	33.1	33.3	32.7	32.5	32.8	32.7	32.0	31.9	31.9	31.4
	5%	WB	25.4	25.6	26.2	27.0	27.4	27.2	27.2	27.2	26.8	27.0	26.7	26.0
		MCDB	31.3	32.0	32.7	33.0	32.3	31.9	32.2	32.0	31.3	31.4	31.3	31.1
	10%	WB	24.8	24.9	25.7	26.5	26.9	26.7	26.7	26.7	26.3	26.5	26.2	25.5
		MCDB	30.6	31.5	31.9	32.2	31.8	31.2	31.5	31.2	30.5	30.7	30.8	30.5
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	11.5	10.9	10.1	9.0	8.4	9.1	8.9	8.1	8.4	9.4	10.9
		MCDBR	12.1	12.0	11.5	10.8	10.0	9.5	10.2	9.9	9.2	9.5	10.2	11.6
		MCWBR	5.8	5.3	4.8	4.3	4.3	4.6	5.3	5.2	4.6	4.8	5.2	5.9
	5% WB	MCDBR	11.6	11.5	10.9	10.3	9.4	9.1	9.8	9.3	8.8	8.9	9.7	10.9
		MCWBR	6.3	5.8	5.4	4.9	4.7	4.8	5.5	5.3	4.7	5.0	5.4	6.2
(40) Clear-Sky Solar Irradiance	taub		0.397	0.423	0.496	0.597	0.639	0.491	0.496	0.484	0.482	0.441	0.416	0.399
	taud		2.446	2.372	2.152	1.884	1.799	2.222	2.216	2.249	2.233	2.379	2.427	2.458
	Ebn at Noon		897	886	829	745	703	807	805	824	829	858	871	885
	Edh at Noon		110	123	157	206	220	143	144	141	143	121	111	107
(44) All-Sky Solar Radiation	RadAvg		5.34	5.87	6.23	6.19	5.42	5.22	5.81	5.72	5.13	4.98	5.19	5.12
	RadStd		0.14	0.21	0.21	0.26	0.36	0.31	0.37	0.33	0.36	0.39	0.21	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page